

3	(a) (i) force ($= mg = 0.15 \times 9.81$) = 1.5 (1.47) N	A1 [1]
	(ii) resultant force (on ball) is zero so normal contact force = weight or the forces are in opposite directions so normal contact force = weight or normal contact force up = weight down	A1 [1]
	(b) (i) (resultant) force proportional/equal to rate of change of momentum	B1 [1]
	(ii) change in momentum = $0.15 \times (6.2 + 2.5)$ (= 1.305 N s)	C1
	magnitude of force = $1.305/0.12$ = 11 (10.9) N	A1
	or	
	(average) acceleration = $(6.2 + 2.5) / 0.12$ (= 72.5 ms^{-2})	(C1)
	magnitude of force = 0.15×72.5 = 11 (10.9) N	(A1)
	(direction of force is) upwards/up	B1 [3]
	(iii) there is a change/gain in momentum of the floor	M1
	this is equal (and opposite) to the change/loss in momentum of the ball so momentum is conserved	A1 [2]
	or	
	change of (total) momentum of <u>ball and floor</u> is zero so momentum is conserved	(M1) (A1)
	or	
	(total) momentum of <u>ball and floor</u> before is equal to the (total) momentum of <u>ball and floor</u> after so momentum is conserved	(M1) (A1)

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